

Track 42: Mechatronics Systems Integration

June 2026

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Departments: Mechatronics · Mechanical · EEE · ECE | MR ROKESH TECHNOLOGY

Skill path: Mechanical Design → Electronics → Embedded Control → PLC → Robotics → System Integration → Industry 4.0

Related: [Track 37 — CAD/CAM](#) | [Track 25 — PLC](#) | [Track 23 — ROS2](#)

Track Overview

Who this is for: Mechatronics students integrating mechanical, electrical, and software subsystems into complete automated systems.

Prerequisites: Basic physics and programming; [Track 22 Arduino](#) or equivalent helpful.

Tools: SolidWorks/Fusion 360, Arduino/STM32, PLC simulator, ROS2, MATLAB/Simulink, pneumatic/electro-pneumatic kits.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Mechatronics awareness	Sensor-actuator integrated project
1 Month	80–100	Junior integrator	Automated workstation model
3 Months	250–300	Internship-ready	PLC + embedded capstone
6 Months	500–600	Mechatronics engineer entry	Industry 4.0 system capstone
9 Months	750–900	Advanced integrator	Autonomous production cell

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Mechatronics system overview, sensors, actuators	System block diagram
D3–4	Arduino/embedded control of motors, relays	Motor control circuit
D5–6	Pneumatics/electro-pneumatics basics	Pneumatic circuit + control
D7–8	PLC ladder integration with sensors	PLC-controlled station
D9–10	Capstone integration	Automated sorting/conveyor station

1-Month Foundation

Week	Topics	Project
W1	Mechanical design basics (CAD intro or review)	Mounting bracket design
W2	Embedded control + sensor fusion	Multi-sensor controller
W3	PLC programming for automation sequence	3-station sequence
W4	HMI + system integration	Integrated mechatronics station

3-Month Job Ready

Month	Topics	Projects
M1	CAD + mechanical assembly design	Designed automation fixture
M2	Embedded + PLC hybrid control	Hybrid control system
M3	ROS2 or robot arm integration, capstone	Smart manufacturing cell capstone

6-Month Professional

M1–M3 + M4: Industry 4.0 (IoT monitoring) + M5: Vision-guided quality check + M6: Full production cell capstone.

9-Month Advanced

Capstones: Autonomous AGV + robotic arm workcell + digital twin documentation | **Level:** L5

Assessment Rubric

Weekly Quiz 20% | Integration labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

This track **integrates** [Track 37](#), [Track 25](#), [Track 23](#), and [Track 19](#). Assign capstone matching institution lab equipment.